

PRODUCT DESIGN ENGINEERING

CardClutch Technical Report

Design Engineering MEng Year 2



Pigeonhold (Team 8)

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1 Product Opportunity [1]

The Problem / Market Opportunity

Archivist, small businesses and makers frequently need to create custom packaging for irregularly sized items. However, existing methods are often not streamlined. Users typically have to measure the object manually, create a suitable box net, transfer the design onto the material, and then cut it out by hand. This makes the process repetitive and tedious, especially when packaging multiple unique items. It can also result in inefficient material use, inconsistent fit, and a reliance on larger or specialist equipment that is often too expensive or impractical for small-scale users. Overall, there is no widely available tool that quickly generates accurate, item-specific packaging without specialist equipment or training, highlighting a clear gap in the market.

Pre-existing Products

The ChompSaw^[2] is a safety-focused cardboard cutting tool designed primarily for educational and maker-space environments. While it improves safety and allows users to quickly cut cardboard, it remains focused only on the cutting stage of the process. Users would still be required to manually measure and design the packaging layout beforehand, meaning the workflow is not automated and still depends on user judgement and accuracy. It also offers limited flexibility, as the user must move the cardboard around the tool rather than the tool around the cardboard. Furthermore, it is relatively expensive and less portable than a handheld alternative.



Low-cost handheld tools such as cordless electric rotary cutters^[3] provide a portable and flexible way to cut materials like cardboard. However, they do not help with measuring, designing, or generating packaging shapes, so the user must do all of this manually. They are also less suited to following box-net paths, as sharp corners and changes in direction are difficult to cut accurately. This is because the circular blade cannot easily pivot on the spot. As a result, accuracy depends entirely on user skill. User testing also showed that it can be difficult to cut in a straight line, with users often moving off the intended path.



Overall, existing solutions are not specifically designed for the creation of bespoke packaging. This creates an opportunity for a more purpose-built solution that reduces effort and better supports small-scale users.

Executive Summary

CardClutch is an all-in-one handheld device that enables users to create and cut custom box nets directly from physical objects. The user first measures the approximate dimensions of an item using the integrated measuring tools, after which the system automatically generates an optimised box net.

A circular display, laser guidance and haptic feedback then guide the user through the cutting process step-by-step, reducing the need for specialist knowledge or experience. The cutting mechanism is enclosed within the device and is manually operated, improving safety while maintaining accuracy. Internal electronics, sensors and feedback systems are arranged using an optimised layout and shielding to minimise interference and ensure reliable operation.

Wider Reach

Although CardClutch is primarily designed for archivist, the need for accurate, custom-fit packaging exists across multiple sectors. For example, archaeological teams may require a portable solution for creating protective storage boxes on-site, while theatres could use the device to quickly produce packaging, props or stage components. In each case, users benefit from a faster process and improved control over the final result. This wider market potential was supported through talking with Rachael Sparks from the UCL institute of Archaeology, as well as feedback from visitors at GERF, who expressed interest in how CardClutch could be applied within their own fields.

CardClutch also improves accessibility by reducing reliance on precise manual measurement and cutting. The guided interface makes the process easier for users with less experience, lower technical confidence or reduced dexterity.

2 Concept Development + Prototyping Strategy [4]

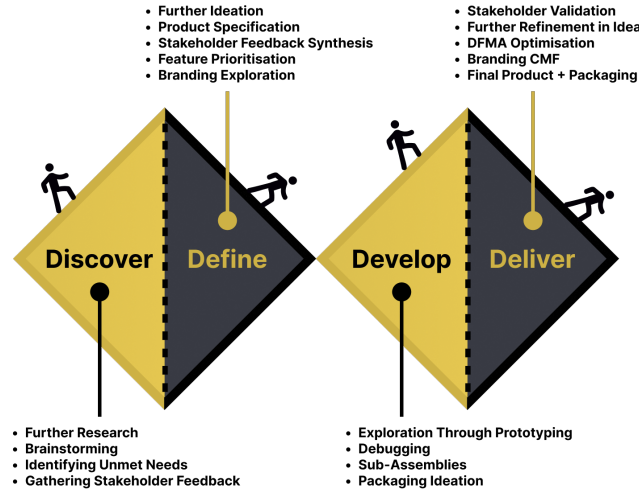


Figure 1: Double Diamond Design Process

The concept development process will follow a Double Diamond framework consisting of four iterative stages: **Discover**, **Define**, **Develop**, and **Deliver**. Although structured, the process will remain non-linear, with feedback from users, tutors, and internal reviews leading to returns to earlier stages when required. This ensures all decisions remain centred around user needs, and product requirements and specifications.

Discover (Research + Opportunity Identification)

User research, market analysis, and technical benchmarking will be used to identify unmet needs and define a clearer opportunity area. Four initial product directions will be evaluated, before being narrowed to a single final concept through tutor feedback, user input, and team evaluation, ensuring alignment with user needs and project feasibility. From this, initial concept features will be explored through brainstorming, sketching, and user surveys, combining stakeholder feedback with design judgement to inform early prototyping decisions.

Define (Specification + Concept Framing)

A weighted decision matrix will first be used to evaluate the initial concept options and feature ideas. The matrix criteria will be defined using Stanley & Scotch brand values to ensure consistency with their product design language, including durability, reliability, intuitiveness, affordability and commercial appeal.

From the results of the decision matrix, a reduced and refined set of features will be selected for further development. A detailed engineering specification will then be created, defining the requirements these features must meet. This specification will form the basis of the prototyping phase.

Develop (Further Ideation + Prototyping)

Selected features will be iteratively developed and refined through sketching, CAD modelling, and subsystem-based prototyping, until all defined criteria are satisfied. This will allow each function to be validated independently prior to full system integration.

The product will be divided into key subsystems: measuring, cutting, and guidance, followed by final system integration. Each subsystem will follow a structured iterative cycle of:

Prototype → Test → User Feedback → Analyse → Refine

The completion of each subsystem will be treated as a key milestone, allowing performance and progress to be evaluated and user feedback to be incorporated into further refinement. Low-fidelity prototypes will be used to validate form, ergonomics, and early mechanical function, while higher-fidelity prototypes will be used to assess electrical performance and system integration. This subsystem-based approach will reduce integration risk by identifying issues early and enabling targeted debugging.

Deliver (Refinement + Finalisation)

Final development will include full system integration, DFMA optimisation, branding and CMF refinement, ensuring compliance implementation, and packaging integration. The final outcome will be a fully validated product and packaging system that meets all functional and user requirements defined in the specification.

3 Branding Consideration & Rationale

3.1 Competitor Analysis

Ryobi^[5] has built strong commercial value through its broad range of home and garden power tools and its ONE+ battery platform, which encourages customers to remain within the brand ecosystem as they add tools over time. This creates repeat purchases while reducing the perceived cost of buying additional products and reduces battery waste. They operate under environmental and safety standards such as WEEE and RoHS. From a user-experience perspective, Ryobi makes power tools feel more accessible to non-experts through clear, approachable product design. This shows how products can stay accessible while still meeting safety and sustainability requirements.



Scotch^[6] creates business value by offering a highly recognisable and trusted range of adhesive products across home, school and workplace settings. It is supported by 3M technology, which ensures consistent adhesion and durability. This allows the brand to maintain a premium, associated with dependable performance and the technical credibility. For users, this reliability reduces uncertainty as they can select a Scotch product with confidence that it will perform as expected. This highlights the importance of strong underlying technology to build brand trust.



Stanley^[7] creates business value through its established reputation for durable, practical tools used across both professional and DIY markets. Its “built to last” positioning supports a premium by assuring customers that products will withstand demanding use and provide long-term value. For users, its familiar and practical designs make robust tools feel accessible to less experienced users. This demonstrates how perceived longevity and reliability can help build brand loyalty across a wider tool ranges.



Swiss Tools creates business value through its premium “100% Swiss Made” positioning, in-house manufacturing and lifetime guarantee, which differentiate the brand in a competitive precision tool market. Reinvesting around one-fifth of turnover into product development helps maintain technical innovation, quality and long-term brand value, while certifications such as ISO 9001 and ISO 14001 reinforce credibility around quality and sustainability. For users, the lifetime guarantee and precision led manufacturing reduce uncertainty when purchasing higher value tools. This shows how quality assurance and after sales confidence can justify a premium while improving user trust.



3.2 Branding Rationale

A collaboration between Scotch and Stanley was selected to develop the CardClutch brand identity. This decision combines two complementary brand strengths and values. Stanley aligns with the functional requirements of CardClutch through its reputation for durability and reliability, alongside its long heritage in toolmaking, making it suitable for the accuracy and control required in an electromechanical product.

Scotch supports the user experience side of the product by positioning itself as an advocate for “do-ing” and “do-ers.” Its straightforward, user-focused communication, such as “No mess, no tools, and seriously strong,” makes products feel accessible and easy to understand. This helps reduce the perceived complexity often associated with industrial equipment, lowering barriers for users with little or no prior experience.

Together, the collaboration strengthens overall market appeal by balancing technical capability with ease of use. Stanley provides credibility and precision, while Scotch ensures accessibility and approachability. This positions CardClutch as a product that is both capable and easy to engage with, supporting a wider and more inclusive user base.

4 Subsystem Prototyping

A weighted decision matrix was used to evaluate early ideas against defined design criteria. These were based on Stanley & Scotch's product design language. The criteria were:

- Durability – resistance to wear and repeated use (/5)
- Reliability – consistent performance over time (/5)
- Intuitiveness – ease of understanding and use (/5)
- Affordability – cost-effective for production and end users (/5)
- Feasibility – engineering practicality and manufacturability (/5)
- Commercial appeal – market desirability and user value (/10)

Commercial appeal was given a higher weighting to ensure a strong business case, mass market applicability, and inclusive design considerations are prioritised within the final concept selection.

4.1 Measurement Subsystem_[8]

This subsystem is responsible for capturing object dimensions and user movement to generate accurate cutting paths. The specification and requirements for the subsystem are shown below.

Key Prototyping Question: How accurately and consistently can the system track user movement and object dimensions across repeated use and different users?

Specifications and requirements

Table 1: Requirements, Specifications, and Justifications for the Measuring Subsystem

Requirement	Specification	Justification
Linear measurement accuracy	± 5 mm accuracy	± 5 mm was chosen because using low-cost sensors keeps the overall system cheaper, making it more accessible to a wider market. Small dimensional errors are also absorbed during folding and assembly of cardboard without affecting overall function. User feedback also indicated that exact accuracy is not essential, provided the box remains usable.
Angular measurement accuracy	$\pm 3^\circ$ accuracy	Angular accuracy is required to ensure that generated cut paths and fold lines remain correctly aligned, particularly at corners and angled surfaces. A tolerance of $\pm 3^\circ$ provides sufficient directional control for box-net generation while avoiding the need for unnecessarily high-precision sensing hardware.
Measurement repeatability	$\geq 80\%$ similarity across repeated measurements of the same object	Repeatability is important because the system should produce consistent packaging outputs when measuring the same object multiple times. An 80% similarity target allows for minor variation from user handling and sensor noise, while still reducing the need for repeated measurements, manual correction, or redesign. This aligns with Stanley's emphasis on reliability.

Track Ball Mechanism

The trackball mechanism was developed to track device movement across the cardboard. An IMU was initially investigated for this purpose, but was found to be unsuitable because it tracks device motion rather than movement relative to the cardboard, reducing reliability. In addition, the introduction of the magnetic clutch mechanism raised EMC compliance concerns, further limiting its suitability. As a result, a trackball mechanism was adopted. Initially, this was created manually using a sphere and two encoders, but it was later redesigned as a compact trackball breakout to reduce assembly size and improve measurement accuracy. By mounting it on the front clamping assembly, the mechanism remains pressed against the cardboard during operation, ensuring consistent surface contact and improving tracking reliability.

Wheel Mechanism

The wheel mechanism was developed to measure the length and width of the object being packaged. Initially, this function was intended to be combined with the trackball mechanism to reduce component count, cost and assembly complexity. However, user testing showed that a dedicated measurement wheel reduced ambiguity, as users associated it more with a measurement tool. This made the measuring process easier to understand, supporting inclusive design for users with varying levels of technical experience. Despite increasing component count, the improvement in intuitiveness was considered more valuable for the intended mass-market audience.

TOF Sensor Mechanism

The ToF sensor was developed to measure object height, completing the dimensions required for automatic box generation. An optical flow sensor was also prototyped and evaluated; however, testing showed it was less accurate and increased overall system cost. The ToF sensor provided more consistent measurements with higher repeatability at lower cost, making it better suited to the project's accuracy, repeatability and affordability requirements. As a result, it was selected as the preferred solution, aligning with Stanley's emphasis on reliability.

Milestone 1

Interviews with Matti from the V&A and Rachael Sparks from the UCL Institute of Archaeology confirmed that the separated measurement functions were more intuitive than earlier concepts, although both highlighted the need for clear on-screen guidance. This reinforced the need to integrate the UI with the measurement subsystem to make the system more accessible to non-technical users. Component testing resulted in the wheel mechanism being reduced in size to improve linear accuracy, as the smaller circumference improved tracking resolution. Furthermore, a stopper was incorporated into the shell design to allow users to stop the wheel at the desired position more easily, improving measurement accuracy and reducing human error. Following these adjustment, all requirements except intuitiveness were achieved, identifying usability as the key area for further improvement in future prototyping.

4.2 Guidance Subsystem [9]

This subsystem provides directional feedback to guide the user along the intended cutting path. The specification and requirements for the subsystem are shown below.

Key Prototyping Question: How effectively do the visual and haptic feedback systems guide the user to follow the correct cutting path with clarity and confidence?

Specifications and requirements

Table 2: Requirements, Specifications, and Justifications for the Guidance Subsystem

Requirement	Specification	Justification
Laser alignment accuracy	± 3 mm lateral deviation (from intended cutting path)	The laser is used for visual guidance rather than measurement, so performance is limited by hand movement rather than system precision. Smaller deviations will also not be easily perceived, so tighter tolerances would not improve usability.
Haptic feedback response time	< 300 ms response time	This ensures haptic feedback feels instantaneous during interaction while accommodating processing and actuation delays within the prototype system, avoiding noticeable delay in user response.
Clarity of guidance cues	User can accurately interpret and describe guidance cues without prompting	This allows guidance and UI clarity to be evaluated through how users naturally interpret the cues, reflecting real-world usability and whether the system can be understood on its own. This reduces the need for user training and supports a simple, easy-to-use system in line with Scotch's emphasis on simplicity.

Haptics

The haptic feedback system was implemented to provide vibration-based corrective feedback when the user deviates from the intended cutting path, offering an immediate non-visual cue. Audio feedback was also investigated, however, to improve inclusivity, it was not used as the primary method in order to avoid reliance on a single sense. Instead, it was implemented as an optional setting to support different user preferences.

Laser Guide

The laser guide was developed as a physical complement to the digital interface, providing a visible projection of the intended cutting path. It reduces cognitive load by providing a direct physical reference in the working environment. An alternative LED-ring system linked to an IMU was also considered, where deviation would be indicated through more LEDs lighting up as the user moved further off path, however, this was rejected due to potential visual distraction, reduced simplicity in use, inclusivity concerns around colour use, and previously identified IMU reliability limitations. The laser was therefore selected as a simpler and more intuitive guidance method, aligning with Scotch's values on, easy-to-use design.

Sat-nav Interface

The satnav-style interface was developed to provide real-time visual guidance during cutting by displaying the intended path and updating the user's position as they progress. Using a familiar navigation-based layout makes the system easier for first-time users to understand and more accessible to users with different levels of technical experience. A dedicated rotary dial was developed alongside the interface, allowing users to adjust the zoom level to suit their preference. This was informed by feedback from Matti at the V&A, who highlighted the importance of understanding the wider context of the cut, rather than only viewing the immediate cutting path. Following a semiotic analysis of an existing market product, knurling was incorporated into the dial to improve grip and provide a clear affordance, making it easier to identify and operate. This reinforces Scotch's emphasis on simple, intuitive user interactions.

Milestone 2

Peer and tutor review of the satnav interface indicated that a purely digital guidance system could be difficult to interpret, especially for first-time users. This validated the inclusion of the laser guide as a physical reference to help alongside the on-screen guidance. Stakeholder interviews and survey feedback also identified haptic feedback as the preferred correction method, as it provided immediate guidance without requiring users to divert visual attention from the cutting task. Feedback also suggested that audio feedback could become annoying over time, leading to having it as an optional setting rather than the primary feedback. Subsequent testing showed that all the requirements were achieved, providing validation of the guidance subsystem.

4.3 Cutting Subsystem [10]

This subsystem enables controlled cutting while maintaining user safety. The specification and requirements for the subsystem are shown below.

Key Prototyping Question: How safely can the blade be deployed, and how accurately can it perform cutting during user operation

Specifications and requirements

Table 3: Requirements, Specifications, and Justifications for the Cutting Subsystem

Requirement	Specification	Justification
Blade coverage	100% blade coverage when not cutting	This ensures the blade is fully enclosed when not in use, removing the risk of accidental contact during handling, transport or setup. Any exposed blade would still present a safety risk, especially in situations where the product is used by individuals with different levels of experience. This would better align with Stanley's emphasis on safe tool design.
Successful cut completion	User completes full cut without blade stalling or material tearing	This was chosen as partial cuts or material damage would prevent the packaging nets from being usable, particularly in contexts such as archives where small tears can affect item protection and condition requirements. A pass/fail approach was therefore used to provide a clear and practical measure of performance.
Piercing success	Blade punctures the cardboard on the first attempt	This tests whether the blade depth and deployment mechanism can initiate a cut. First-attempt puncture is important, as repeated attempts suggests insufficient cutting force or poor alignment, reducing usability and increasing task time. This aligns with Stanley's focus on reliable, precise tool performance.

Retractable Blade Mechanism

The retractable blade mechanism was made to control vertical blade deployment using a lead screw driven system. The original concept involved a motor-driven rotary cutting blade, as this is the approach used by many existing cardboard cutters, including the benchmark product evaluated during testing. However, this approach restricted the ability to pivot during cutting due to the guided rail system used to support the material, preventing the creation of more complex shapes such as packaging nets. As a result, the design was changed to a stationary scalpel-style blade, allowing manual directional control and pivoting. This also reduced system complexity and improved safety by removing continuous blade rotation. The final solution therefore provided greater cutting flexibility while maintaining the simple, user-focused interaction associated with the Scotch × Stanley brand identity.

Motor Calculation + Justifications [11]

The actuation mechanism converts rotational motion from a stepper motor into linear motion using an M6 lead screw with a 1 mm pitch. Cardboard failure is assumed to occur through localised shear and fibre rupture at the scalpel tip. The cutting force is therefore estimated using an effective shear-based model:

$$F \approx \tau \cdot (t \cdot L) \approx 1.2 \times (6 \times 0.5) = 7.2 \text{ N}$$

Where L = effective engaged cutting length at scalpel tip (mm), t = cardboard thickness (m) To account for variability in material density, blade wear, misalignment, and dynamic effects during puncture, a safety factor of 5 is applied:

$$F_{design} = 7.2 \times 5 = 36 \text{ N}$$

The axial force is converted into lead screw torque, with a conservative correction factor of 3 applied to account for stiction, rail misalignment, non-uniform puncture resistance, and transient impact loads during initial penetration.

$$\tau_{real} = 3 \left[\frac{F \cdot p}{2\pi\eta} \right] = 3 \left[\frac{36 \times 0.001}{2\pi \times 0.35} \right] \approx 0.049 \text{ Nm}$$

Where p = M6 lead screw pitch(m), η = Conservative mechanical efficiency A 1:1 gear ratio was used to directly couple the motor to the lead screw within the compact casing, with small transmission losses accounted for in the torque model:

$$\tau_{motor} = \frac{\tau_{real}}{\eta_g} = \frac{0.049}{0.85} = 0.058 \text{ Nm}$$

Based on the required torque of 0.058 N·m and the need for a low-cost, widely available motor suitable for rapid prototyping, a NEMA 17 stepper motor was selected. It also enables multiple operating modes, including blade retracted (no exposure), blade pivoting, and two discrete cutting depth levels. The selected motor has a holding torque of approximately 0.45 N·m, with a conservative usable torque of 0.15–0.25 N·m during operation. This provides a safety margin of roughly 2.5×–4×, ensuring the motor can reliably handle the required load, including the initial puncture, while maintaining stable operation across all modes.

Blade Clamp Mechanism

The blade clamp mechanism was designed to hold the scalpel blade in position while allowing it to be replaced once it becomes worn. Easy blade replacement was considered important to extend the lifespan of the product and reduce the need to replace the entire cutting assembly. It also encourages users to replace the blade regularly, as a sharper blade improves safety by reducing the force required to cut. A screw-based clamping system was selected in which the blade is secured by tightening a screw, providing a simple and reliable solution while still allowing straightforward removal. An alternative blade holder was also considered, where the blade is inserted and secured using a screw thread mechanism that compresses the holder around it. However, this proved difficult to integrate within the available space and offered less flexibility for accommodating different blade types, limiting its suitability for wider application. The screw-based solution was therefore more practical and easier to implement for both prototyping and longer-term use.

Magnetic Card Clutch Mechanism

The magnetic card clutch mechanism was developed to hold the cardboard firmly in place before cutting begins. It uses two magnetic plates to clamp the material between the upper and lower surfaces, preventing movement during cutting and improving stability and accuracy. This was introduced as a safety feature following the change to a stationary exposed scalpel blade. The magnetic clutch acts as an interlock, only allowing the blade to deploy when the material is correctly positioned and secured. This improves safety, particularly for users with no technical experience, while keeping the mechanism simple to operate. Although the use of magnets

introduces potential EMC considerations, this was accepted as a trade-off for improved usability and safety. This approach also aligns with Stanley’s emphasis on accessible tool design by ensuring the system remains safe and stable for non-professional users.

Milestone 3

Milestone 3 evaluation was conducted using tutor feedback, user interviews, and internal prototyping tests to validate the cutting subsystem. A key outcome was validation of the magnetic card clutch mechanism, which was confirmed as an effective solution for stabilising the material during cutting and removing the need for a separate cutting mat. One user, however, noted a preference for using a cutting mat for comfort, leading to this being implemented as an optional feature. This also allows for more consistent cutting conditions across different users and contexts, particularly for those with limited access to additional equipment. Initial prototyping identified issues with blade actuation, as the original motor lacked sufficient torque and blade deployment was slower than anticipated. These issues were addressed by refining the motor calculations, selecting a higher-torque motor within budget constraints, resulting in improved reliability. Different gear ratios were also tested to find an appropriate balance between blade deployment speed and sufficient torque. Following iterative development and validation, the cutting subsystem achieved all set requirements, confirming the viability of the final design for safe and accessible use.

5 Colour, Material, Finish [12]

5.1 Colour Considerations

A black and yellow colour scheme was deliberately selected to align with Stanley’s established brand identity and existing product range. This ensures immediate visual recognition as a Stanley product and supports seamless integration into their wider portfolio, reinforcing key brand values such as durability, reliability and professional quality. While the colour scheme is primarily driven by Stanley’s identity, it also aligns with Scotch’s use of bright, often primary, high-contrast colours. This helps create a visually cohesive approach across both brand influences

Beyond branding, the colour scheme was also developed with consideration for usability and accessibility. The high-contrast combination of black and yellow improves feature legibility and supports quick identification of interaction points on the product, which is especially useful in fast-paced or low-focus contexts. Limiting the palette to two contrasting colours also improves accessibility for users with colour vision deficiencies. The colour scheme remains distinguishable for the majority of users with colour blindness, with reduced clarity mainly affecting those with tritanopia, which accounts for less than 1% of the population [13]. This was supported through user testing on showcasing the prototype, where a visitor with partial colour blindness reported a preference for the Stanley-inspired concept. They noted that it made key features easier to distinguish compared to competitor products and found the overall design more visually appealing.

5.2 Material and Manufacturing Rationale

A stiffness-to-weight performance index ($E^{1/2}/\rho$, where E is Young’s modulus and ρ is density), based on bending-dominated structural optimisation, was used to compare the structural efficiency of materials (Ashby and Johnson [14]). This focuses on maximising stiffness for minimum mass, which is critical for maintaining structural performance without increasing weight. Material cost was not included in the final selection stage, as it is relatively small compared to internal component costs and has a reduced impact when considered at bulk manufacturing scale.

Table 4: Material selection comparison for Main Casing

Material	Young’s Modulus (Pa)	Density (kg/m ³)	Performance Index
ABS	2.42×10^9	1045	47.1
Polycarbonate	2.38×10^9	1200	40.7
Nylon	1.49×10^9	1135	34.0
Polypropylene	9.22×10^8	902	33.7
30% Glass-filled Nylon	5.00×10^9	1290	54.8

Note: Material property values are approximate and based on Granta EduPack data or datasheet.

Although 30% glass-filled nylon is more expensive than the other materials considered, it offered the highest stiffness-to-weight performance and the greatest resistance to deformation. The additional cost was considered acceptable due to the improved durability and rigidity, particularly as the casing represents only a small proportion of the overall product cost. Its use is also consistent with the robust, long-lasting qualities associated with Stanley products.

Injection moulding was selected for the main casing as it is well suited to producing 30% glass-filled nylon parts at high volume with consistent quality. During processing, the molten material in the mould cavity aligns the glass fibres in the direction of flow, helping achieve the required stiffness and structural performance. The process also provides good dimensional accuracy and repeatability, ensuring consistent part fit during assembly, which is important for mass-market production. Although glass-filled nylon can lead to issues such as shrinkage and warping due to fibre orientation, these effects can be controlled through careful tooling design and process settings, such as maintaining uniform wall thicknesses and adjusting gate positions to control how the material enters and fills the mould. Despite this, injection moulding remains suitable for producing complex geometries with consistent quality at mass scale.

An overmoulded layer was then introduced to improve ergonomics in areas of direct user contact, especially due to the rigidity of glass-filled nylon. This creates a softer outer surface while maintaining the structural integrity of the core, improving grip and comfort during repeated handling. Additional overmoulded sections were also included at the magnetic clutch interface, both on the base and the upper clutch surface, to improve grip and make it easier for users to separate the two parts.

To select the overmould material, an Ashby plot was used to compare perceived softness ($S = E \times H$, where E is Young's modulus and H is Vickers hardness), as defined in Ashby's material selection methodology [14], against tensile strength. Perceived softness relates to how soft or compliant the material feels on contact, affecting user comfort in gripping areas. Tensile strength represents resistance to tearing and permanent deformation under repeated handling. This allows materials to be compared based on comfort and durability when selecting suitable overmould options, supporting Stanley's focus on long-term durability and performance.

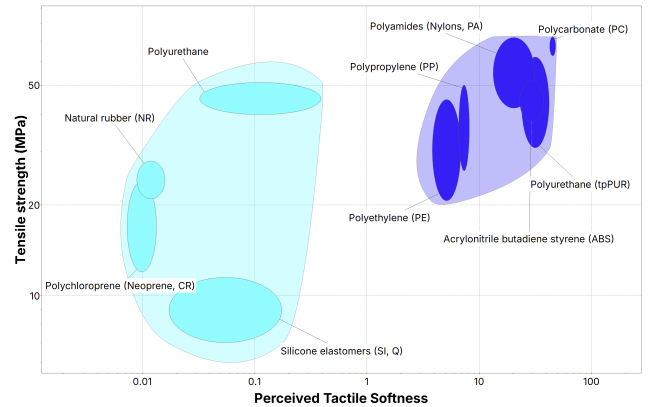


Figure 2: Ashby Plot Comparison

Based on this, a polyurethane elastomer was selected as it provides a good balance of perceived softness and tensile strength compared to the other materials considered. It provides a softer surface for improved user comfort while still resisting tearing during repeated use.

5.3 Finish

The product finish was developed through a combination of market analysis, ergonomic considerations, and manufacturing requirements. A semiotic analysis of competitor products identified textured surfaces as a common visual cue for grip and user interaction areas. This feature is also frequently used in Stanley products, supporting brand continuity and making the intended handling areas more intuitive. In production, the grip pattern could be integrated directly through overmoulding. However, as the current prototype uses 3D printing for the overmoulded parts, adding this texture would increase manufacturing complexity, so it is better suited to a future production model. Knurling was added to the adjustment dial to improve grip while also acting as a semantic cue to show its a rotational interaction and making it more intuitive.

Manufacturing considerations also influenced the final finish. The visible exterior surfaces of the casing were positioned against the mould cavity to achieve a smoother, higher-quality finish, while less visible internal surfaces had functional moulding features such as ejector pin marks. Parting line locations were positioned along the upper and lower edges of the product, where they are concealed by the display housing and overmoulded grip features. This minimises the visibility of surface defects on the main user-facing areas, reducing the amount of post-processing required. The parting line also follows the natural split between the upper and lower casing halves, meaning the mould can be split into two main halves without requiring more

complex features such as side actions. During prototyping, boss geometries were also refined and thickened where necessary to reduce the likelihood of visible knit lines forming around the screw locations used to assemble the casing.

6 System Integration

Table 5: Requirements, Specifications, and Justifications for the Guidance Subsystem

Requirement	Specification	Justification
System usability	User completes a guided cut without requiring external assistance.	This was used to see whether the product could be used independently without assistance. If a user requires help, it suggests that part of the guidance, interface, or workflow is unclear. Successfully completing a guided cut means the system is intuitive to use, making it accessible to users with different levels of experience and more suitable for a mass-market product.
Operational comfort	User does not adjust grip or reposition their hand more than twice during a cut	This provides a measurable way to test ergonomics during use. If the user needs to adjust their grip or move their hand more than twice during a cut, it means the product is uncomfortable, unstable, or difficult to control. Keeping adjustments to two or fewer shows that the user can hold and use the system comfortably throughout the whole workflow.
Subsystem compatibility	All major functions operate without preventing the use of another function.	This ensures that combining subsystems does not cause issues during operation and that they can work without interfering with each other. If this requirement is not met, it would mean there are integration problems such as mechanical obstruction (i.e. between gears and wiring) or parts of the system failing when multiple functions are running at the same time.

6.1 Main Casing [15]

DFM/DFA/Injection Moulding Consideration

The casing design was developed in alignment with ISO 8887 principles for design for assembly and disassembly [16], with a focus on reducing part count, simplifying assembly, and supporting end-of-life separation where possible.

Uniform wall thickness was maintained wherever possible to ensure consistent cooling during injection moulding, reducing the risk of sink marks, warping, and internal stress development. This also makes the way molten plastic fills the mould cavity more predictable, improving consistency between manufactured parts, which is desirable for mass manufacturing. Internal and external corners were rounded with radii between $0.6\text{--}0.75 \times$ wall thickness to reduce stress concentrations and improve part ejection by removing sharp corners that could increase resistance during demoulding. This also helps prevent weak points and reduces the likelihood of incomplete filling during manufacture.

Draft angles of approximately 1° were applied to all vertical faces (0.5° draft on ribs) to make ejection from the mould easier and reduce wear on tooling. This is important for repeatable high-volume manufacturing, as it reduces part sticking and improves tool lifespan. Rib features were used to increase stiffness without increasing overall wall thickness, improving the case's structural performance without significantly increasing mass, and maintaining suitability for injection moulding. Rib thickness was kept at 1.5 mm, equivalent to 53% of the main wall thickness, keeping it within the recommended 50–75% range to reduce the risk of sink marks from localised thick sections.

Bosses were designed to provide secure mounting points for fasteners while remaining compatible with injection moulding requirements. Buttress ribs were used to avoid excessive material thickness in one area, which could lead to sink marks or uneven cooling during manufacture. Counterbores and clearance holes were included to ensure screws do not bottom out and to reduce stress around the screw location. 20mm Thread-forming M3 screws were selected to provide sufficient pull-out strength in thermoplastic material for repeated assembly and disassembly, supporting easier maintenance of the product.

The estimated thread stripping force_[17] was calculated as:

$$P = \frac{1}{2} \pi \cdot d \cdot L \cdot \tau = \frac{1}{2} \pi (3)(20) \left(\frac{139}{\sqrt{3}} \right) \approx 7.6 \text{ kN}$$

This value is significantly greater than any loads expected during use, including potential impacts, confirming that the 20 mm engagement length provides sufficient retention for the application. The use of standardised M3 screws also simplifies sourcing and replacement, while supporting ISO 8887 principles by reducing fastener variety and overall part count.

Chamfers and tapers were incorporated to aid both injection moulding and assembly. In moulding, they help guide molten plastic into tighter regions of the tool and make it easier for the part to be released after cooling. During assembly, chamfered edges help parts guide themselves into position, so they don't need to be perfectly aligned beforehand. This reduces the chance of incorrect assembly and makes the process more time efficient.

Additional self-locating and self-guiding features were integrated into the casing to reduce alignment effort during assembly. For example, the motor is located in a tight-fitting slot that holds it in position, and small locating pins are used to align components such as the limit switches through their mounting holes. This means parts naturally sit in the correct position during assembly, rather than needing to be carefully positioned by hand or held in place while being fitted. This reduces assembly time and improves consistency between builds, which is important for mass-manufactured products. Clearance was also included around key areas to ensure components can be easily placed without needing additional tools such as tweezers.

In line with ISO 8887 guidance, the design avoids over-constraining parts by using only the minimum number of locating features needed to position components accurately. This reduces the build-up of manufacturing tolerances between parts, so small differences in the size of each component do not combine and affect how well the parts fit together during assembly. This was achieved by giving each internal component its own dedicated rib or slots. The two halves of the shell were aligned using protruding bosses with counterbored interfaces, helping the parts self-align during assembly. As the shell is asymmetrical, the correct orientation is also visually obvious, reducing the risk of incorrect assembly. Symmetry was used where possible to reduce orientation errors, while asymmetric features were used where necessary so parts can only be assembled in the correct way. This applies Poka-Yoke (mistake-proofing) principles by making incorrect assembly difficult or immediately obvious. Overall, this improves repeatability between builds, making the product more suitable for mass production and reducing waste caused by assembly errors.

Snap Fit Considerations

The snap-fit was designed as a cantilever arm for glass-filled nylon. An arm length and width of 16 mm and 6 mm were selected to allow elastic deflection during assembly while maintaining structural stability. Keeping the width below half the arm length prevents excessive stiffness, reducing insertion force and risk of brittle failure in glass-filled nylon.

A 1.8° taper was used to gradually reduce stiffness along the arm, spreading bending more evenly along its length rather than concentrating it at the root, while maintaining strength at the fixed end.

A 0.7 mm fillet was included at the base of the cantilever to reduce stress concentration in the region of highest bending stress, lowering the risk of crack initiation during assembly and potential disassembly.

A hook depth of 0.6 mm was initially selected, however, prototyping showed it was not secure enough with the mating feature. The depth was therefore increased to 1.2 mm to improve retention while limiting the deflection required from the snap arm.

A 1.0 mm hook length was used to spread the load during locking, reducing local stress at the contact area.

A 30° ramp angle reduces insertion force by allowing the hook to slide gradually into position during assembly, while a 75° retention face prevents reverse movement after assembly, ensuring a secure lock.

To validate the design, an FEA was carried out to determine the stresses in the snap-fit during insertion and compare them against the tensile yield strength of the material (139 MPa). The assembly force was first estimated using cantilever beam theory:

Assembly Force Calculation_[18]

$$F = \frac{3EIy}{L^3} = \frac{3(6823)(4)(1.2)}{16^3} = \frac{98251.2}{4096} \approx 24.0 \text{ N}$$

Applying a 25% safety factor to account for real-world conditions:

$$F_{\text{FEA}} = 1.25 \times 24.0 = 30.0 \text{ N}$$

This load was used in the finite element analysis.

FEA Results

$$\text{Max Stress} = 46.388 \text{ MPa} < 139 \text{ MPa}$$

This confirms the snap-fits will operate within the elastic region with no permanent deformation under assembly loading.

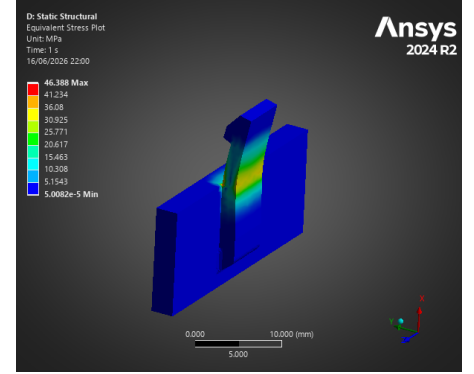


Figure 3: FEA Stress Plot for Snap Fits

6.2 Design Iterations

Several design changes were made to improve assembly, reduce part count, and increase structural performance. The laser module was redesigned so it partially protrudes from the casing, improving internal space efficiency and preventing interference with the shell, which previously obstructed part of the laser. The lead screw rail system was integrated directly into the casing rather than being a separate part, reducing part count and simplifying assembly by removing the need for component alignment during assembly. Snap-fit features were reviewed as they would require side actions for mould ejection due to undercuts, however, they were kept to reduce the number of fasteners required and simplify overall assembly. To optimise rib placement and validate structural performance, a stress analysis and frequency study were carried out.

The stress analysis identified the highest stresses at the transition between the handle and the front housing, although these remained below the tensile yield strength of glass-filled nylon (139 MPa). Buttress ribs were therefore used in these higher-load areas, as their tapered geometry provides a more gradual load transfer into the surrounding casing compared to standard ribs, reducing bending at the rib roots and increasing local stiffness.

After this, a frequency study was carried out to ensure the casing would not resonate with the motor excitation frequency. The motor operating frequency_[19] was calculated as:

$$f = \text{RPS} \times \left(\frac{360^\circ}{\theta_{\text{step}}} \cdot N_{\text{microstep}} \right) = 2 \times \left(\frac{360}{1.8} \times 1 \right) = 400 \text{ Hz}$$

The fundamental frequency of the structure was found to be 211 Hz, providing sufficient separation from the motor operating frequency and reducing the likelihood of resonance. This helps prevent excessive vibration, improving both product reliability and user comfort. Furthermore, the natural frequency is well outside the hand-arm vibration range identified in ISO 5349-1_[20] (10.9–17.3 Hz), which is associated with potential joint injury in the wrist, elbow, and shoulder, reducing the likelihood of vibration-related discomfort during use_[21].

6.3 Battery Rationale and Power Analysis

The system is powered by a single lithium-ion cell, with a 15 V USB-C PD supply used during prototyping for safety. A dual-rail architecture is used, with a 15 V rail for the stepper motor and a 5 V rail for control electronics. This separation is used because the motor has high transient power demand, while the control system requires stable low-voltage power. To estimate battery demand, the system is analysed over a typical operating cycle. The device does not run at full load continuously, so a 4-minute cycle is used: 1 minute of motor operation for blade deployment and retraction, and 3 minutes of low-power sensing and control.

5 V & Stepper Motor Consumption

The logic system (microcontroller, sensors, display, feedback) runs continuously, so its energy use is calculated over the full cycle.

$$\text{Power, } P = V \cdot I = 5 \times 0.6 = 3 \text{ W}$$

$$\text{Energy per Cycle, } E = P \cdot t = 3 \times \frac{60}{4} = 0.20 \text{ Wh}$$

The same method is applied to the stepper motor, giving an energy consumption of 0.22 Wh per cycle.

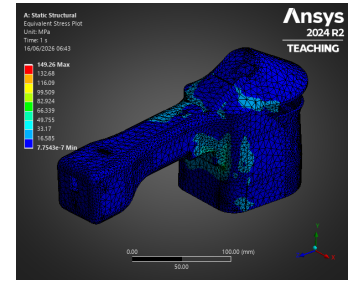


Figure 4: FEA Stress Analysis for Casing

Total energy per cycle

All energy contributions are combined to give the total cycle consumption for the defined functional unit, which is one complete cutting cycle (one box). A 15% loss factor is included to account for inefficiencies in the buck converter, motor driver circuits, and other electrical losses such as heat generation. The system is designed to achieve a minimum of 20 functional units per charge, which defines the minimum energy requirement for battery selection.

$$E_{cycle} = [(0.20 + 0.22) \times 1.15] \times 20 \approx 9.66 \text{ Wh}$$

6.3.1 Battery Selection

To determine the battery requirement, the cycle energy is converted into an equivalent lithium-ion capacity using the nominal cell voltage of 3.7 V:

$$\text{Ah} = \frac{9.66}{3.7} \approx 2.60 \text{ Ah}$$

However, this is a theoretical minimum and must be increased in practice to account for voltage drop under load, reduced usable capacity at high discharge rates, battery ageing, and the minimum operating voltage of the boost converter. To account for these factors, the usable capacity is assumed to be approximately 65% of the rated cell capacity, giving a practical value of ~ 4.00 Ah. Therefore, A Molicel P42A (4.2 Ah, 45 A continuous rating [22]) was selected, as it exceeds the calculated practical capacity requirement while remaining within a suitable single-cell 21700 form factor.

Peak Current and C-Rate Validation

The selected battery must also be capable of supplying the peak current demanded by the system during motor actuation. Using the total peak system power of ~ 39.6 W, the maximum battery current is estimated as:

$$I_{cell} = \frac{P_{load}}{V_{cell} \cdot \eta} = \frac{39.6}{3.7 \times 0.85} \approx 12.6 \text{ A}$$

where $\eta = 0.85$ accounts for conversion losses and inefficiencies in the power delivery system. For the selected 4.2 Ah cell, the corresponding C-rate[23] is:

$$C\text{-rate} = \frac{12.6}{4.2} \approx 3C$$

The Molicel P42A is rated for a continuous discharge current of 45 A, equivalent to approximately 11C. As the required discharge rate is only 3C, the battery operates well within its capability, providing substantial electrical and thermal headroom while reducing voltage sag during operation, validating the battery selection.

Further Validation + Safety Constraints

The system is limited by the minimum input voltage required by the boost converter supplying the 15 V rail, which must remain above approximately 3.0 V under load to maintain stable output. During high-current operation, the battery may experience voltage sag towards this threshold, and cells with higher internal resistance may drop below the usable operating range even when charge remains. For this reason, a high-discharge lithium-ion cell was required to maintain stable voltage during peak loading.

6.4 User Interface [24]

The user interface was divided into 5 main sections: Measure, Generate Box, Cutting Space, Vector Upload, and Settings.

The Measure function combines data from the measuring wheel and time-of-flight sensor to determine object dimensions, removing the need for manual measurement and improving consistency. These dimensions can then be transferred directly to the **Generate Box** function, where a box net is automatically generated, reducing the number of user steps required. For users who already know the dimensions of an object, measurements can also be entered manually, allowing the measurement stage to be skipped entirely.

The Cutting Space function provides real-time cutting guidance using a sat-nav style interface. Rather than requiring users to draw out a box net themselves, the system tracks progress and indicates when a turn is required. This is shown through on-screen guidance, where a progress bar gradually fills as the user rotates to the correct angle. During turning, the blade is partially retracted to allow safe pivoting before cutting resumes. This reduces cognitive load and improves cutting accuracy, particularly for inexperienced users.

The Vector Upload function allows users to store and load box nets or custom cutting files from an

SD card. This enables repeat production of standardised packaging while also extending the product's functionality beyond only packaging applications.

The Settings menu allows users to customise feedback methods, including enabling or disabling audio and haptic feedback, and setting whether the magnetic clutch mechanism is required for blade deployment. Units of measurement can also be adjusted, allowing the system to operate in either metric or imperial formats depending on user preference. This improves accessibility by allowing the product to accommodate different user needs and working environments without requiring any changes to the hardware.

7 Final Assembly

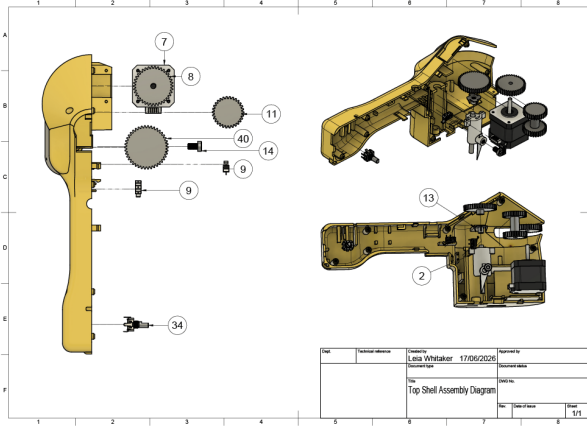


Figure 5: First Half of Assembly

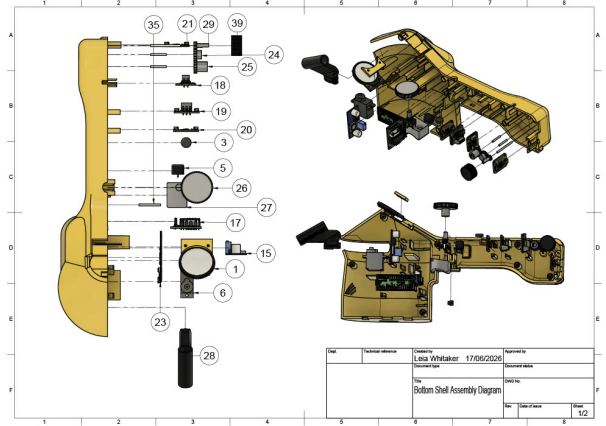


Figure 6: Second Half of Assembly

The CardClutch assembly follows ISO 8887 principles to reduce part count and simplify construction through self-locating geometry and modular subassemblies. This eliminates the need for adhesives and minimises the use of mechanical fasteners, which are only used for securing the main housing. Standardised M3 fasteners are used where required to maintain consistency and reduce assembly variation. Overall, the design prioritises ease of assembly, serviceability, and reduced build complexity.

To evaluate assembly efficiency, a Boothroyd-Dewhurst DFMA index was calculated using the relationship:

$$E_{ma} = \frac{N_{min} \cdot t_a}{t_{ma}}$$

Where t_a = theoretical min assembly time per part (assumed as 3 seconds), N_{min} = theoretical min no. of parts, t_{ma} = total assembly time.

A teardown analysis of a benchmark rotary cutter identified approximately 12 parts. As the product was already highly simplified, this was taken as the theoretical minimum part count. Using an estimated assembly time of 65 seconds from the Boothroyd-Dewhurst tables, $E_{ma} \approx 55\%$.

CardClutch contains approximately 44 parts in its current form. However, several of these components could be integrated more effectively in future iterations. For example, multiple electronic modules could be combined onto a single PCB, reducing both part count and assembly operations. Taking these opportunities into account, the theoretical minimum part count was estimated at approximately 30 parts. Using an estimated assembly time of 210 seconds, $E_{ma} \approx 43\%$.

The DFA efficiency shows that, although CardClutch is more complex than a simple rotary cutter, it is still relatively efficient to assemble. The increase in functionality does not lead to a large increase in assembly effort, meaning the design remains practical to manufacture and can still be improved further in future iterations.

8 Product Compliance

As CardClutch is a battery-powered electronic cutting device intended for small-business use, it must comply with UK product safety, electrical, environmental and waste regulations. Hence, compliance considerations were integrated throughout the design process to ensure the final product would be suitable for releasing to the public.

Table 6: Summary of Relevant Product Compliance Requirements & Implementations

Regulation / Standard	Relevance to CardClutch	Design Implementation
General Product Safety Regulations 2005 ^[25]	Safe under intended use and foreseeable misuse	Fully enclosed cutting mechanism, activation prevention unless correct conditions are met, laser guidance and haptic feedback to reduce user error.
Electrical Equipment (Safety) Regulations 2016 ^[26]	Prevent electric shock, fire and overheating	Low-voltage lithium-ion system, internal isolation, circuit layout protects against overcurrent and thermal failure.
EMC Regulations 2016 / BS EN IEC 61000 ^[27]	Prevent electromagnetic interference	Shielding and layout optimisation for motors, sensors, haptics and laser systems.
Restriction of Hazardous Substances (RoHS) Regulations 2012 ^[28]	Restriction of hazardous substances	Hazardous substances such as lead, mercury and cadmium not used in electronic components.
Waste Electrical and Electronic Equipment (WEEE) Regulations 2013 ^[29]	End-of-life recycling and disposal	Partial disassembly design, disposal guidance included, WEEE symbol applied.
IEC 62133 / UN 38.3 / UN 3481 ^[30]	Lithium battery safety and transport	Lithium-ion battery with protection against overcharge, overcurrent and overheating.

Before being released to market, CardClutch would require UKCA and CE marking to demonstrate compliance with relevant UK and EU regulations. A UK Declaration of Conformity would need to be issued, confirming that the product meets all applicable safety and compliance requirements^[31]. A technical file would also be produced and maintained for 10 years, containing key documentation such as risk assessments, FMEA analyses^[32], electrical schematics, safety and EMC test results, user instructions and compliance records.

8.1 Safety Validation and Consumer Considerations

Both intended use and foreseeable misuse were considered during the development of CardClutch. Key hazards included accidental contact with the blade, unintended activation, battery failure and electronic interference. These risks were reduced through a fully enclosed cutting mechanism, controlled activation conditions, battery protection circuitry and shielding of electronic systems. As users may not fully assess safety risks before operating a product, features such as laser guidance, haptic feedback and enclosed cutting were incorporated to reduce user error and improve accessibility for users with different levels of experience.

9 Product & Package Labelling

CardClutch have been designed to align with regulatory, safety, and environmental requirements for a portable electronic tool intended for UK and EU markets, consistent with established manufacturers such as Stanley. The product includes CE and UKCA markings, indicating compliance with relevant safety, electromagnetic compatibility, and electrical standards required for legal sale in both markets. These are positioned clearly on the device to ensure long-term visibility. The product also includes model identification, serial/batch coding, and electrical rating information to ensure full traceability throughout its lifecycle. Internal structural components are manufactured from glass-filled nylon and marked with the material identification code PA66-GF30. This allows accurate material identification for recycling and engineering traceability.

As an electronic device containing a rechargeable lithium battery, the WEEE symbol is included to indicate that the product must not be disposed of in general household waste and should be recycled through appropriate electronic waste channels. A lithium-ion battery safety label is also applied to highlight safe handling, charging, and disposal requirements due to the risks associated. In addition, a laser warning label is included to inform users that the integrated guidance system uses a Class 2 laser, ensuring appropriate safety awareness during use. A “refer to instruction manual before use” symbol is also included on both the product and packaging to encourage users to read the provided safety and operating instructions prior to use. This is a standard safety communication used on electronic and handheld power tools, particularly those involving moving components and integrated guidance systems, helping to reduce misuse.

For packaging, CE, UKCA, and WEEE markings are included alongside a widely recycled label and FSC certification mark, supporting the sustainable disposal and responsible sourcing of packaging materials.

The packaging is designed using primarily recyclable cardboard materials with minimal non-recyclable elements to reduce environmental impact. It also includes key consumer information such as product identification, barcode, and handling guidance (i.e. keep dry), laid out in a way that does not interfere with branding or key product messaging, maintaining a clear visual hierarchy similar to that used by brands such as Stanley or Scotch.

10 Packaging & User Guide [33]

10.1 Requirements & Specifications

Table 7: Packaging Requirements, Specifications & Considerations

Requirement	Specification	Justification
Product protection	Product remains secure during transport within packaging	The packaging must withstand typical shipping, storage and retail handling conditions to ensure the product arrives fully functional. Any damage to the cutting mechanism would prevent safe use and result in product failure at first use, especially in non-professional environments where users expect the product to be ready out of the box.
Sustainable material use	At least 90% of packaging is made from recyclable cardboard	This was selected as it gives a practical balance between sustainability and structural performance. A 100% target is often unrealistic due to the need for adhesives, coatings and printed layers, while lower values would require more virgin material and increase environmental impact. This also supports Stanley’s wider sustainability target of a 42% reduction in Scope 1 and Scope 2 greenhouse gas emissions by 2030 (from a 2022 baseline). [34]
Safety and compliance information	All relevant safety symbols, warnings, and disposal information are clearly presented on both the packaging and within the user guide.	This ensures that critical safety information is communicated before first use and reinforced during operation. Clear communication of safety and disposal requirements supports compliance and reduces the risk of misuse, particularly for a mass-market user group with limited experience of cutting tools.
Branding consistency	The packaging and user guide both use Stanley × Scotch branding	Consistent branding strengthens product recognition and makes it clear that the packaging and user guide are all part of the same system. As a dual-brand Scotch × Stanley product, consistent CMF helps users instinctively link it to durability and ease of use.

10.2 CMF Considerations

The CMF strategy for the packaging was developed using in-store research of existing Stanley products, where packaging was colour-picked and photographed as reference. This meant the design directly matched Stanley’s packaging style and maintained consistency in visual language, resulting in immediate brand recognition. The packaging material was selected as corrugated cardboard due to its high strength-to-weight ratio, suitability for mass production, and ease of recycling within standard waste systems. It was also chosen to match the material used in existing Stanley packaging, maintaining consistency with the brand’s current product range. From a manufacturing perspective, the outer surface would be printed using flexographic printing due to its suitability for high-volume production. It allows brand colours and graphics to be applied consistently while keeping production costs low. A matte finish was then selected to reduce glare and create a low-gloss, professional appearance aligned with Stanley packaging. Labelling was positioned on the rear face of the packaging next to the barcode to separate compliance information from the main product visuals. This keeps the front focused on the product while still ensuring the required information is easy to locate when needed. A white background was used for the labelling area to improve contrast against the yellow and black packaging, ensuring all information is easy to read and quick to identify.

10.3 Packaging Type Rationale

After evaluating three common packaging styles, the mailer-style box was selected for further development. While the hard-shell case provided the highest level of protection, user feedback indicated that this level of durability was unnecessary for the intended use case, as the product is aimed at homes, archives, and small-scale packaging environments rather than construction sites. Compared to the hard-shell case, the mailer design offers lower manufacturing cost, reduced material usage, and improved recyclability while still providing sufficient protection during shipping and retail handling. The regular slotted corrugated box was also considered; however, the mailer-style option was preferred due to its simpler and more satisfying opening experience. To ensure product stability during transport,

the internal negative space is filled with folded cardboard inserts that secure the product and prevent movement within the box. In addition, the product is enclosed in a protective plastic bag to reduce the risk of water entering the packaging and improve resistance to minor water exposure during handling and shipping.

10.4 Life Cycle Considerations

The packaging was designed with its full life cycle in mind, including shipping, retail use and end-of-life disposal. During shipping, the corrugated cardboard structure provides enough protection while keeping the packaging lightweight, helping to reduce transport costs and emissions. In retail, the simple front-facing layout makes the product easy to identify without needing any additional display packaging. After use, the packaging can be fully recycled through normal household waste streams. It can also be kept and used to store the product and any accessories, rather than being thrown away immediately after opening.

10.5 User Guide [33]

The user guide was designed to follow the same structure used in established Stanley hand tool documentation as part of the Scotch x Stanley branding, keeping it consistent with familiar industry formats. It begins with technical data and regulatory information, including CE marking and manufacturer details, to confirm compliance and give key product context upfront.

Safety content is prioritised using danger, warning, and caution categories, shown in larger bold text and supported by the standard warning symbol. This clear hierarchy makes higher-risk information immediately distinguishable and reduces the chance of misinterpretation. Key operational hazards are reinforced with simple icon-based instructions covering risks such as battery handling, water exposure, and contact with conductive materials.

Additional information includes package contents and a labelled component overview to reduce setup errors by helping users correctly identify parts. Intended use is clearly defined with explicit “DO NOT” restrictions to prevent unsafe operation. The guide concludes with assembly and adjustment instructions, supported by images, to ensure correct initial setup of the product.

11 Final Design [35]

CardClutch is a handheld assistive card-cutting tool designed primarily for home users, archives, and small-scale packaging environments, combining measurement, box generation, and cutting into a single device. A measuring wheel and Time-of-Flight sensor are used to capture object dimensions, while a screen interface and control knob allow users to navigate menus, generate box templates, and adjust settings. A trackball tracks user movement and enables a screen-based navigation system that guides cutting paths using real-time sensor input. A laser provides a visual reference during cutting operations to improve accuracy. The blade is deployed using a dedicated trigger and protected by a magnetic blade cover to improve safety when not in use. An ergonomic grip supports comfortable prolonged use, while a rechargeable battery system accessed via a USB-C charging port enables portable operation.

The approximate product cost was estimated using supplier prices for purchased components and estimates for custom parts, giving a total unit cost of approximately £115 at single-unit production. In an industrial context, this would reduce significantly due to economies of scale and bulk purchasing.

11.1 Design Validation [36]

An injection moulding simulation was carried out in Fusion 360 to assess whether the casing could be manufactured successfully and to identify any potential moulding defects before production. Both casing halves were analysed using 30% glass-filled PA66 with a melt temperature of 290°C and a mould temperature of 80°C.

The results showed that both casing halves can be moulded successfully, with 89–93% of the geometry classified as easy to fill. Both parts filled in approximately 2 seconds and cooled in 3.6–4.1 seconds, indicating that the material can flow throughout the mould without difficulty. Recommended mould shrinkage allowances were between 0.852% and 0.939%, while packing pressures remained below 80 MPa, suggesting that no unusually high moulding pressures are required. No sink marks were predicted on one casing half and only a single sink mark was identified on the other, indicating good control of wall thickness throughout the design. Seven weld lines were predicted across both halves, however these were located away from highly stressed or critical features and are therefore unlikely to affect structural performance. The transition between the handle and front housing was identified as the most challenging region to manufacture. This area exhibited the greatest shrinkage, longest filling times, and deepest sink mark formation due to the change in section thickness between the handle and housing. As a result, this region was identified as the primary area for future design optimisation.

12 Analysis of Prototyping and Testing

The prototyping process followed a progressive iteration approach, starting with low-fidelity subsystem prototypes before refining them and moving towards full system integration. This allowed individual functions to be developed and tested in isolation before being combined into the final system. Early testing used simple unit tests, including quick physical mock-ups such as a wheel mechanism linked to a rotary encoder to check that motion tracking would be accurate. Software-based unit tests were also written and used to validate that sensor data, such as time-of-flight measurements, could be correctly captured and processed. This helped confirm that both the physical and digital components would work as intended before integration. However, system-level testing revealed difficulties when integrating all components into the compact casing.

Different manufacturing methods were used at different stages of the prototype to improve efficiency and speed up iteration. Laser cutting was used early on where possible, as it was significantly faster than 3D printing and allowed basic structures to be tested quickly, especially when determining a suitable gear ratio for the motor. Foam was also used during early form development to explore ergonomics, for example when defining the overall shape of the product and testing how users interacted with the handle. It allowed rapid adjustments while still giving a more realistic sense of form than cardboard. As the design progressed, 3D printing was used for later iterations due to the increased complexity of the geometry and the need to test part fit, assembly, and overall form factor, including design-for-manufacture considerations. In some cases, only small sections were printed, such as snap-fit tests, to validate specific features without printing the full assembly. This reduced material use and iteration time while still allowing key design decisions to be tested. It also provided a more realistic representation of the final product, including the use of coloured filament to better communicate appearance.

Testing used a balance of functional testing and user testing. User testing was based on short, task-based trials to observe how users interacted with the system and its ergonomics in a more realistic use case. Functional testing focused more on the technical performance of the system, helping identify issues such as timing delays, tracking errors, and mechanical problems, including instances where the trackball began to slip due to low friction. Using both approaches allowed the system to be evaluated from both a user and technical perspective. However, user testing was mainly observational and qualitative, which made it more difficult to gather quantitative data and directly compare results across different iterations, especially when different users were involved in later tests.

A key issue in the early prototyping stage was that some system constraints were underestimated, especially actuation force for the scalpel blade, and software library compatibility. In later iterations, this was improved by identifying potential issues earlier and planning design changes in advance where possible. For example, design-for-manufacture improvements were made to the CAD model while the previous iterations were still printing, allowing integration testing to begin immediately once parts were available. Potential issues were also managed using an MVP (Minimum Viable Product), value add, and nice-to-have prioritisation structure. This helped focus effort on the most critical features first and ensured that important subsystems were developed and tested early, reducing the risk of them becoming bottlenecks later in the process. This prioritisation was used to define a clear critical path, focused on completing MVP functionality before moving on to secondary features. User feedback directly influenced this path, with insights from testing being fed into decisions about which features were prioritised or developed further. This ensured that the most important user requirements were addressed early rather than left to later iterations.

12.1 Further Development Recommendations

Custom PCB Design

A custom PCB could be developed to replace the loose soldered wiring and perfboards used in the prototype. This would reduce the space required for electronics, improve reliability, simplify assembly, and make the product easier to manufacture at a larger scale. However, it would have needed to be designed several weeks in advance to allow time for manufacturing, testing, and potential redesigns if issues were identified.

Optimising Geartrain & Motor Combination

Testing identified the blade actuation motor as one of the main contributors to the overall product size. Further development could therefore explore alternative gear ratios and smaller motors to achieve the required actuation force while reducing both product size and component cost. However, this could introduce a trade-off, potentially reducing the speed of blade actuation depending on the gear ratio used.

Increased Modularity

The internal structure could be redesigned to be more modular by splitting the product into physical sections (front, middle and head), rather than separating it into functional subsystems such as guidance, measurement and cutting. Each section would contain a combination of functions, but remain independently accessible. This would make faults easier to locate within the product and simplify maintenance and repair, as individual sections could be removed or replaced without affecting the rest of the system.

Automatic Adjustment for Material Thickness

The cutting subsystem currently operates using four depth settings. Future development could use automatic adjustment based on cardboard thickness, allowing a continuous range of depths rather than fixed steps. This would make the system suitable for a wider range of cardboard thicknesses and reduce the likelihood of incomplete cuts.

Improved Design for Manufacture

Although many manufacturing considerations were implemented in the final design, further improvements could be made by making the wall thickness in the internal structure more consistent to better suit injection moulding. This would reduce the risk of sink marks, warping, and uneven cooling during production.

Hall Effect Sensor for Magnetic Clutch Detection

Due to project time constraints, development focused primarily on minimum viable product functionality. Future versions could incorporate a Hall effect sensor to detect when the magnetic clutch is engaged, adding an additional safety feature by preventing the blade from deploying unless the system is active. This could also be linked to the UI as a toggle option, allowing users to enable or disable the feature if they wish to use a cutting mat.

13 Project Planning Discussion

13.1 Gantt Chart Reflection [37]

The Gantt chart was used throughout the project to plan and monitor progress, but it was adjusted several times as development progressed. A progression tracking section was included within the plan and updated regularly, making the chart a live document of the project. This allowed the team to see how up to date each task was, identify when work was falling behind schedule, and adjust priorities where needed. It also highlighted which areas of the project were complete and which still required development.

Milestones were used as checkpoints to measure progress and keep the team aligned. These were placed at the end of key stages, such as completing individual subsystems, as well as at key project events like user feedback sessions during the Discover phase and preparation for the Great Exhibition Road Festival. Regular team meetings were held twice a week (Monday and Thursday) to review progress and ensure tasks were being completed in line with the plan. Between meetings, short focused sprints were often used to respond to tight deadlines and unblock dependent work. However, some issues took longer to resolve than planned. At the start of the project, progress was slowed by a bottleneck while we waited for survey results to help narrow down which features should be prioritised for prototyping. A further challenge emerged during market research, where we found that many existing products were limited to straight-line cutting and did not allow the blade to pivot during use. This made them unsuitable for producing box nets effectively, so additional time was needed to explore alternative mechanisms before selecting a final approach.

CAD modelling also took longer than expected, as each major design iteration had to be manufactured, assembled and tested before the next version could be developed. Due to print complexity and manufacturing time, each iteration created at least two days of downtime, which delayed several tasks that depended on completed prototypes. The planned final completion date was initially set for 15th June to allow time for final review and submission preparation. However, this date was adjusted multiple times due to missed internal deadlines and additional iterations being required beyond what was originally expected.

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